(17)

254 02468

RAMAKRISHNA MISSION VIDYAMANDIRA
(A Residential Autonomous College affiliated to University of Calcutta)

Admit Roll Number
261
Invigilator's Signature with Date : 27/2

1) Show the state space representation of the 8-puzzle problem.

→ for a 8-puzzle problem to represent the state space we have to define 6 things.

(I) states (II) initial states (III) action (IV) Transition model (V) goal test (VI) path cost.

here, (I) states are the diffienet positions of the blank spot in the puzzle. the blank spot can be in 9 places.

(II) initial state

the initial problem state of this puzzel suppose we have this state,

1	2	3
	4	6
7	5	8

. and from this state we have to solve it.

(III) action.

on the basic of this problem, there are total 4 actions we can perform in this puzzel. left, right, up, Down.

(IV) Transition model.

for each action what will be the next state this is the Transition model.

like, if S is the current state and on the basis of this state if we perform a action the the transition model will be RESULT (S, a).

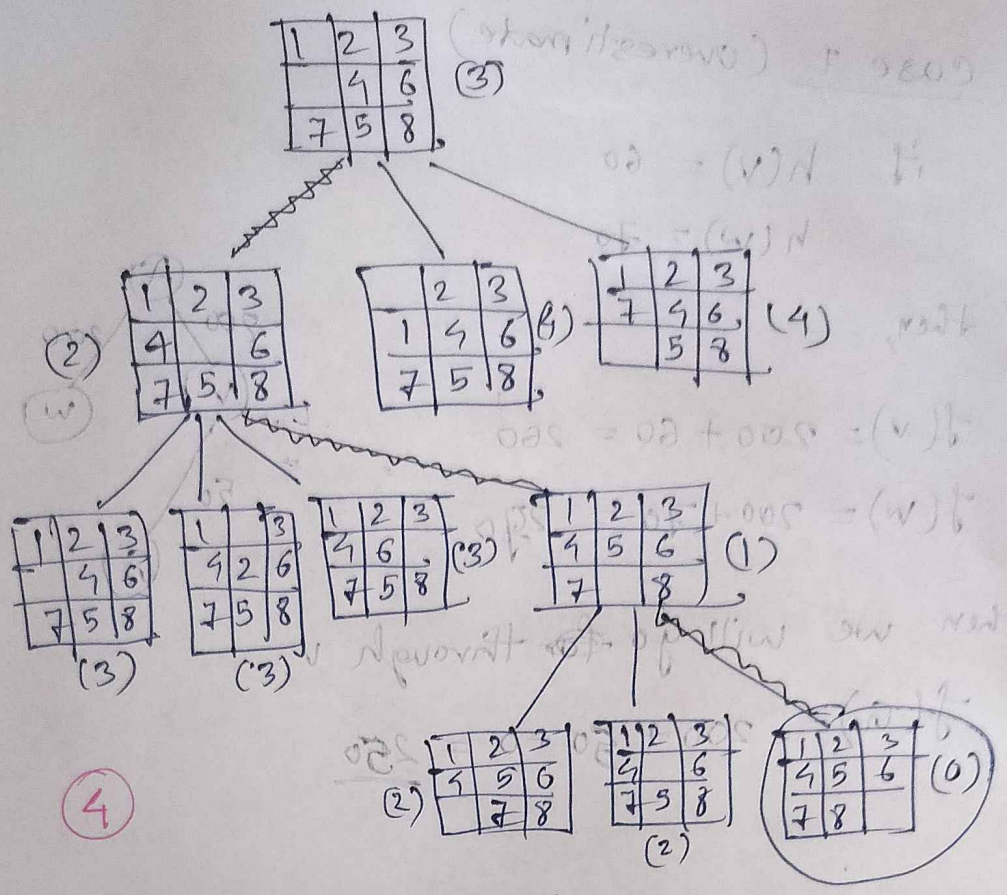
(V) Goal Test

goal test means the solution state. after solving the puzzel ~~will~~ what will be the goal state. That is

1	2	3
4	5	6
7	8	

(VI) Path cost

For each movement ~~will~~ what will the cost, that is called step cost and total step cost is called path cost.

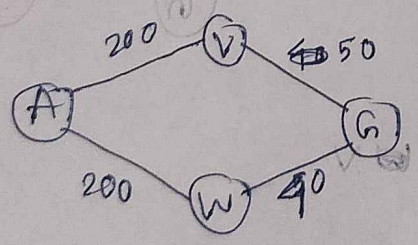


4

② Prove that A^* is admissible.

→ A^* is admissible means there should be at least one optimal solution for this algorithm.

Proof:



~~h(v) =~~

~~Whe~~

here, $h(n)$ means heuristic value

and $h^*(n)$ mean actual value

and, if $h(n) \geq h^*(n)$ that means overestimate

if $h(n) \leq h^*(n)$ that means underestimate.

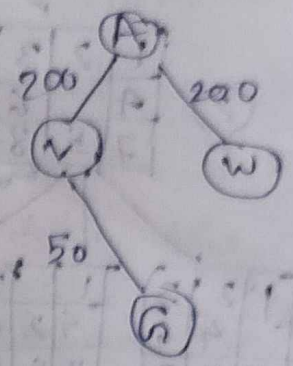
~~Case I~~

Case I (overestimate)

if $h(v) = 60$

$h(w) = 70$

then,



$f(v) = 200 + 60 = 260$

$f(w) = 200 + 70 = 270$

then we will go ~~to~~ through v,

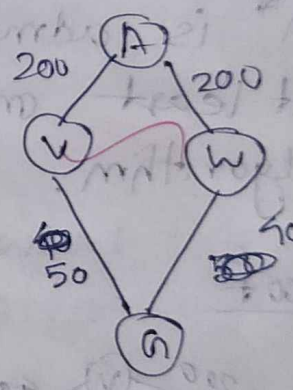
$f(G) = 200 + 50 + 0 = \underline{250}$

Case II (underestimate)

if $h(v) = 20$, $h^*(v) = 50$

$h(w) = 30$, $h^*(w) = 40$

then,



$f(w) = 200 + 30 = 230$

$f(v) = 200 + 20 = 220$

~~when~~ ~~when~~

then we will go through v

then $f(G) = 200 + 50 = \cancel{240}\ 250$

this is not optimal as we have 230

~~so,~~ then we will go through w,

then $f(G) = 200 + 40 = 240$.

So, we get an optimal value $\underline{240}$

3

RAMAKRISHNA MISSION VIDYAMANDIRA
(A Residential Autonomous College affiliated to University of Calcutta)

254 02473

Admit Roll Number
261
Invigilator's Signature with Date:

(5) For a ~~task~~ environment,

PEAS Denots
① Performance measure
② Environment
③ Actuators.
④ Sensors.

(a) playing a single player card game.

① performance measure
cost effective,
win situation,
less loss.

② Environment
the oponent,
cards.

③ Actuators
through cards,
sorting the cards

④ Sensors
eyes/camera,

(2)

(b) Finding landmines in battlefield.

(1) performance measure

successfully cross the battlefield without stepping onto any of the mines.

(2) Environment:

grasses, bushes, tree, rough land, landmines,

(3) Actuators.

move front, back, left, right, defuser landmines, wheels.

(4) Sensors.

eyes / camera, metal detector.

(3) When would Best First Search be worse than Breadth First search? Explain.

→ Breadth First search is performed by expanding its successors. Level by level after executing all the nodes in the same level then we go to the next level for execution and finding goals.

for, Best first search, we move forward on the basis of heuristic value. if there are multiple successors then we look for

the least heuristic values path to find the goal.

now, when Best first search be worse than Bredth first search.

(I) Breadth first search actually determins the shortes path to reach the goal but Best first search does not give us the shortes path. The solution path given by the Best first search would cost more than the optimal path.

(II) If there is a circle, Breadth first search can handel this. But for Best first search it can not.

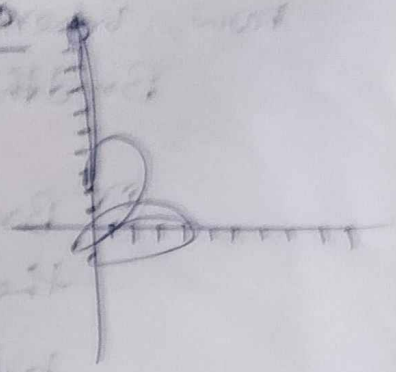
(3)

(III) for same heuristic values path, Best first would be confused to choose the best path. But in case of Breadth first search, it executes all the nodes it have it does not relay on heuristic values.

(4) $f(x) = -x^2 + 6x + 10$

(a) if the hill climbing search ~~starts~~ from $x=0$,

then $f(x) = -0 + 0 + 10 = \underline{10}$



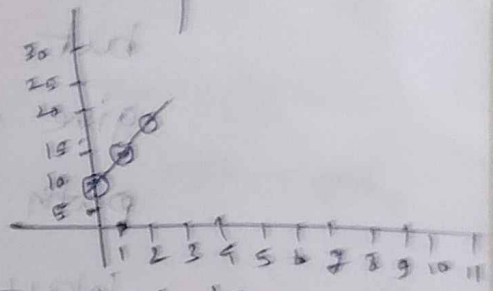
for, $x = 1$

$f(x) = -1 + 6 + 10 = \underline{15}$

for, $x = 2$

$f(x) = -4 + 12 + 10$

$= \underline{18}$



(1)

it is getting bigger, that means the algorithm moves towards the maximum.

(b) if the algo starts from $x = 10$

$f(x) = -100 + 60 + 10$

$= \underline{-30}$

$x = 11$

$f(x) = -121 + 66 + 10$

$= \underline{-45}$

$$\begin{array}{r} 121 \\ -76 \\ \hline 45 \end{array}$$

(1)

it is getting smaller, that meay it is going to the local minimum.

it does not reach the global maximum.